

ABSTRACT

Rollopod - A Transforming Robot with Dual-Mode Locomotion

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Overview

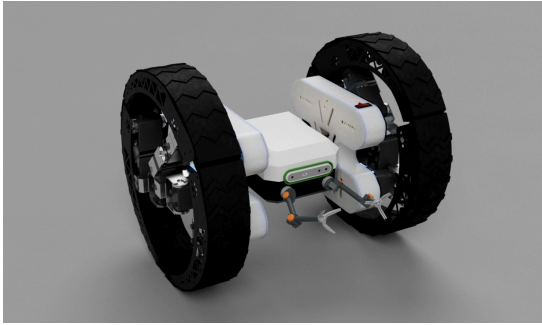


Fig.1: Rolling State

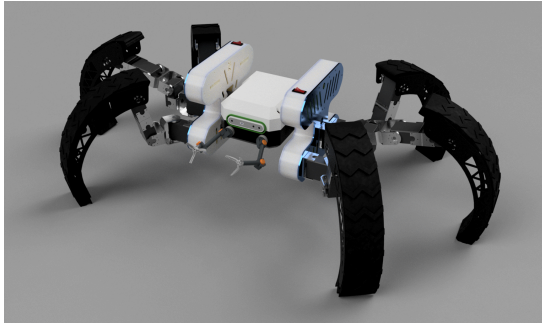


Fig.2: Walking State

Rollopod is a transformable hybrid mobile robot designed to bridge the gap between terrain-adaptive hexapod walking and efficient rolling locomotion. Rollopod reconfigures its six articulated legs into two structural rolling rings, enabling seamless transitions between walking and rolling while maintaining a suspended central body for payload integration and environmental sensing. The project establishes the mechanical, electronic, and software foundation for a scalable robotic platform intended for autonomous field applications.

Objective

The primary objective of Rollopod is to overcome the limitations of existing mobile robots by combining the strengths of both walking and rolling locomotion. Conventional hexapods can traverse rough and uneven terrain effectively but are often limited by their low travel speed. In contrast, wheeled robots can move efficiently over smooth surfaces but struggle in complex environments. Rollopod is designed to bridge this gap by using a single transformable mechanical structure that can switch between walking and rolling without relying on dedicated external wheel assemblies or additional rolling mechanisms. Beyond improving mobility, the project also aims to establish a modular robotic platform that can support future autonomous navigation, environmental perception, and field-assistance applications.

Methodology

The development of Rollopod follows a systematic engineering approach consisting of three primary stages:

1. Mechanical Design

The robot was designed and optimized using CAD to develop a transformable six-legged structure capable of transitioning between walking appendages and dual rolling rings without the need for dedicated external wheels. The design also incorporates a suspended central chassis for payload integration and sensor mounting.

2. Electronics & Control System

A distributed electronics architecture was developed with isolated power distribution to separate high-current actuator loads from sensitive logic circuits. The control system employs a master–slave microcontroller topology, where the master processes commands and the slave synchronizes multi-channel PWM signals for coordinated servo actuation.

3. Software Integration & Validation

Firmware development focuses on coordinated locomotion, transformation sequencing, and modular control for future autonomous operation. Physical prototype assembly and subsystem integration are currently underway to validate the complete design.

(See Ideation Document [1] for full methodology.)

Results

The proposed transformable locomotion concept has been successfully validated through multiple prototype iterations. An initial full-scale **3D-printed prototype** was developed to verify the robot's geometry, transformation mechanism, and overall structural configuration, followed by successive aluminium and steel-based prototypes that improved structural rigidity and mechanical reliability [2].

Experimental testing successfully demonstrated the **rolling locomotion mode** and a preliminary **walking gait**, confirming the feasibility of the proposed mechanical configuration and actuator coordination [2][3]. During walking trials, the failure of a plastic gearbox in the DC motor highlighted the need for higher-strength drive components, leading to subsequent actuator and structural upgrades documented in the project development process [2].

The latest prototype incorporates redesigned powder-coated steel legs, upgraded servo actuators, and an improved mechanical framework, while the electronics, power distribution system, and centralized control module are currently under integration [2]. Comprehensive evaluation of locomotion efficiency, transformation repeatability, and autonomous control will be carried out after completion of the fully integrated prototype.

Conclusion

Rollopod demonstrates that a single transformable robotic platform can combine the terrain adaptability of a hexapod with the efficiency of rolling locomotion without relying on dedicated external wheel assemblies. The prototype development and testing have validated the practicality of this concept while providing valuable insights into structural design, actuator selection, and hybrid locomotion. These findings establish Rollopod as a promising foundation for future autonomous robotic systems capable of operating efficiently across diverse environments, with potential applications in search and rescue, industrial inspection, and field exploration.

Supporting Documents

[1] [Rollopod Ideation Document](#) – Complete engineering documentation covering the concept, design methodology, mechanical architecture, electronics, software, and future development roadmap.

[2] [Rollopod Progress Report](#) – Documents the prototype evolution, fabrication process, design improvements, experimental testing, and current implementation status.

[3] [Rollopod Demonstration Video](#) – Demonstrates the rolling locomotion, preliminary walking gait, and overall prototype development progress.